

SAP S/4HANA — Production Planning and Execution (PP)

Global Bike Case Study

Overview

This project documents a complete, end-to-end manufacturing planning and execution cycle in **SAP S/4HANA 2022 (Fiori 3.0)**, based on the Global Bike dataset. It covers the full flow from material master setup through demand planning, MRP, production execution, and cost settlement.

Role played: Production Manager / Plant Manager / Shop Floor Worker / Goods Receipt Clerk / Warehouse Employee / Controller (multiple roles across the Global Bike Group, as required by each step).

Product: Deluxe Touring Bike (red) — DXTR3###

Process Flow

[PP] Change Material Master → Change Routing → Display Product Group
→ Create Sales & Operations Plan (SOP) → Transfer SOP to Demand Mgmt
→ Run MPS with MRP → Review Stock/Requirements List
→ Convert Planned Order into Production Order

[MM] Receive Goods in Inventory → Issue Goods to Production Order

[PP] Review Production Order Status → Confirm Production Completion
→ Review Production Order Status → Receive Goods from Production Order

[CO] Review Costs Assigned to Production Order → Settle Costs of Production Order

Steps & Results

Replace the placeholders below with your own screenshots and tables from each step.

Step 1 — Change Material Master Record

Prepared the material master records (MRP Data, Forecasting parameters) for the three Deluxe Touring Bikes (black/silver/red) to enable demand planning.

- Strategy Group: 40 (Planning with final assembly)
- Forecast smoothing factors: Alpha 0.20, Beta 0.10, Gamma 0.30, Delta 0.30

Products (3)

Image	Description / ID	Group / Type	GTIN	Product Category	Last Changed
	Deluxe Touring Bike (black) DXTR1400	Finished Bikes (BIKES) Finished Product (FERT)		Product	08/23/2021, 12:36:25 Chris Reich
	Deluxe Touring Bike (silver) DXTR2400	Finished Bikes (BIKES) Finished Product (FERT)		Product	08/23/2021, 12:37:10 Chris Reich
	Deluxe Touring Bike (red) DXTR3400	Finished Bikes (BIKES) Finished Product (FERT)		Product	08/23/2021, 12:37:48 Chris Reich

Deluxe Touring Bike (red)

DXTR3400

Edit

Copy

Show in Hierarchy

Open In

Product Type: Finished Product (FERT)

Product Category: Product

Product Group: Finished Bikes (BIKES)

Base Unit of Measure: Each (EA)

GTIN: --

GTIN Category: --

Revision Level:

General Information

Product Compliance

Components

Configuration

Sales

Storage

Warehouse Management

Extended Service Parts Planning

Distribution Chains

PL

Basic Data

Division:
Bicycles (BI)

Old Product Number:
--

Batch Management Required:
No

Marked for Deletion:
No

Created By:
[Chris Reich](#)

Created On:
08/19/2021, 11:46:23

Last Changed By:
[Chris Reich](#)

Last Changed On:
08/23/2021, 12:37:48

Deluxe Touring Bike (red) /

Plant Dallas

DL00

Product Type: Finished Product (FERT)

Product Category: Product

Product Group: Finished Bikes (BIKES)

Base Unit of Measure: Each (EA)

GTIN: --

GTIN Category: --

Reference Product: --

Fiscal Year of Current Period: 2023

Current Period: 8

General Information

International Trade

Sales

Quality Management

MRP Data

Advanced Planning

Extended Service Parts Planning

Forecasting

General Data

Plant:
Plant Dallas (DL00)

Plant-Specific Product Status:

Valid From Date for Status:

MM/dd/yyyy

Profit Center:

Unit of Issue:

Batch Management Required:
No

Serial Number Profile:

Negative Stocks Allowed in Plant:

Stock Determination Group:

Is Excise Tax Relevant:

Marked for Deletion:

Min. Temperature:

0.00

Max. Temperature:

0.00

Step 2 — Change Routing

Assigned components to operations in the routing for the red Deluxe Touring Bike.

Component Operation

...

...

Operation Overview										
	Op...	SOP	Work Ce...	Plant	* C...	Standard...	Description	Lo...	PRT	CL...
☐	0010		ASSY1000	DL00	ASSY		Material staging	☐	☐	☐
☐	0020		ASSY1000	DL00	ASSY		Attach seat to frame	☐	☐	☐
☐	0030		ASSY1000	DL00	ASSY		Attach handle bar assembly	☐	☐	☐
☐	0040		ASSY1000	DL00	ASSY		Attach derailleur gear assm. to wheel	☐	☐	☐
☐	0050		ASSY1000	DL00	ASSY		Attach front and rear wheels to chain	☐	☐	☐
☐	0060		ASSY1000	DL00	ASSY		Attach brakes	☐	☐	☐
☐	0070		ASSY1000	DL00	ASSY		Attach peddles	☐	☐	☐
☐	0080		INSP1000	DL00	ASSY		Test bike	☐	☐	☐
☐	0090		PACK1000	DL00	ASSY		Disassemble	☐	☐	☐
☐	0100		PACK1000	DL00	ASSY		Pack bike	☐	☐	☐
☐	0110		PACK1000	DL00	ASSY		Move to storage	☐	☐	☐

Item Overview												
	Ph...	Le...	Path	Item...	Component	Quantity	Sort String	Un...	It...	Ba...	Activity Seq.	Cic Material Description
☐	☐	0	0	0010	TRWA1400	2		EA	L			Touring Aluminum Wheel Assembly
☒	☐	0	0	0020	TRFR3400	1		EA	L			Touring Frame-Red
☐	☐	0	0	0030	DGAM1400	1		EA	L			Derailleur Gear Assembly
☒	☐	0	0	0040	TRSK1400	1		EA	L			Touring Seat Kit
☐	☐	0	0	0050	TRHB1400	1		EA	L			Touring Handle Bar
☐	☐	0	0	0060	PEDL1400	1						Pedal Assembly
☐	☐	0	0	0070	CHAN1400	1		EA	L			Chain
☐	☐	0	0	0080	BRKT1400	1		EA	L			Brake Kit
☐	☐	0	0	0090	WDOC1400	1		EA	L			Warranty Document
☐	☐	0	0	0100	PCKG1400	1		EA	L			Packaging

Reviewed the product group PG-DXTR### and the proportion split across the three bike variants.

Name of the product group (1)

[Search and Select](#)

D: Find product group via MRP controller Go Hide Filters ⌵

MRP Controller: Material description: Language Key: Product group: Plant:

⌵ Find Find next Add to Favorites ⌵ ⌵

Items (5)	MRP Contr...	Material description	Language...	Product gr...	P...
☒		400 PRODUCT GROUP BICYCLES	EN	PG-BIKE400	DL00
☐		400 PRODUCT GROUP DELUXE TOURING BICYCLE	EN	PG-DXTR400	DL00
☐		400 PRODUCT GROUP OFFROAD BICYCLES	EN	PG-ORBK400	DL00
☐		400 PRODUCT GROUP PROFESSIONAL TOURING	EN	PG-PRTR400	DL00
☐		400 PRODUCT GROUP TOURING	EN	PG-TRBK400	DL00

Generated a 12-month forecast using historical consumption data (Seasonal Trend Model), then created a production plan matched to target days' supply (5 days).

Period	Sales	Production	Stock Level
...	...	...	...

Forecast: Model Selection



Periods

☒ Period intervals

Forecast

* Fr... : 08/2023

* To: 07/2024

Historical data

* Fr... : 06/2017

* To: 03/2021

☐ No. of periods

No. of forecast periods: 0

No. of historical values: 60

Forecast execution

☐ Constant models☐ Seasonal models☐ Trend models☐ Season. Trend Models☒ Aut. model selection☐ Historical

Forecast parameters

Profile: SAP

[Forecasting](#) [Historical...](#) [Forecast profile...](#) [Version...](#)

Forecast: Results

Basic value: 320.895
MAD: 17Trend value: 6
Error total: 73

Forecast results

Period	Orig. HV	Corr. HV	Ex-post FV	Orig. FV	Corr. FV	Season	F	C
M 04/2021				318	318	0.97	☐	☐
M 05/2021				343	343	1.03	☐	☐
M 06/2021				299	299	0.89	☐	☐
M 07/2021				314	314	0.91	☐	☐
M 08/2021				348	348	1.00	☐	☐
M 09/2021				382	382	1.08	☐	☐
M 10/2021				352	352	0.98	☐	☐

Time series characteristic changed

[Forecasting](#) [User exit](#)

SOP: plan individual product group

Planning Table	Un	M 08/2023	M 09/2023	M 10/2023	M 11/2023	M 12/2023	M 01/2024	M 02/2024
☐ Sales	EA	483	527	484	471	534	582	536
☐ Production	EA	483	527	484	471	534	582	536
☐ Stock level	EA							
☐ Target stock level	EA							
☐ Range of Coverage								
☐ Target days' supply		5	5	5	5	5	5	5

Step 5 — Transfer SOP to Demand Management

Disaggregated the SOP down to the three individual materials (proportional split: DXTR1### 40%, DXTR2### 30%, DXTR3### 30%).

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SAP

Plnd Ind. Reqmts: Planning Table

Search In: "Apps"

Menu

Compare

Delete

History

Planning start: 08/02/2023

Planning End:

Table

Items

Schedule Lines

<<

<

>

>>

	Material	MRP ...	V	A	BU	Req. Segment	M 08/2023	M 09/2023	M 10/2023	M 11/2023	M 12/2023
☐	DXTR1400	DL00	AG	☒	EA		224	214	190	188	216

Planned independent requirements generated per material

Step 6 — Run MPS with MRP

Executed MRP (NETCH processing key) for DXTR3###, generating planned orders and dependent requirements across the BOM.

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SAP

Single-Item, Multi-Level

Search

Menu ▾

* Material: DXTR3400

MRP Area:

Plant: DL00

Scope of Planning

☐ Product group

MRP Control Parameters

* Processing Key: NETCH

* Create Purchase Req.: 2

* SA Deliv. Sched. Lines: 3

* Create MRP List: 1

* Planning mode: 1

Net Change in Total Horizon

Purchase requisitions in opening period

Schedule lines

MRP list

Adapt planning data (normal mode)

* Scheduling: 1

Determination of Basic Dates for Planned

Process Control Parameters

☐ Also Plan Unchanged Components
☐ Display Results Prior to Saving
☒ Display material list
☐ Simulation mode

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SAP

Single-Item, Multi-Level

Menu ▾  Materials

Statistics	
Materials planned	17
Materials with New Exceptions	17
Materials with Termination MRP List	

Parameters	
MRP Area	DL00
Plnt	DL00
Processing Key	NETCH
Create Purchase Requisition	2
SA Schedule Line	3
Create MRP List	1
Planning Mode	1
Scheduling	1

Step 7 — Review Stock/Requirements List

Reviewed the dynamic stock/requirements list, confirming no on-hand stock and reviewing pegged requirements for the first planned order.

Find Order:	000000425	Make-to-stock	Order End Date:	10/01/2023	GR pr.time:	0
Order Qty.:	142	EA	Order Start:	09/27/2023	Proc. Type:	E
Scrap:	0		Opening Date:	09/26/2023	Order Type:	LA

✓ 63 → Prod.Ord. → PartConvProdOrder → Proc.Ord. → SubProcOrd → Pur.Reg. ★ ✕

< SAP
Production order Create: Header

Menu ▾

 Material
 Capacity
 Operations
 Components
 Documents
 Sequences

Order: %0000000001

Type: PP01

Material: DXTR3400

Deluxe Touring Bike (red)

Plant: DL00

Status: REL MACM SETC

General
Assignment
Goods Receipt
Control
Dates/Quantities
Master Data
Long Text
Administration
Items
Fast Entry

Quantities

* Total Qty: 142 EA

Scrap Portion: %

Delivered:

Short/Exc. Rcpt:

Material: DXTR3400
 Description: Deluxe Touring Bike (red)

Ex. manuf.:

MRP Area: DL00

Plant Dallas

Plant: DL00

MRP type: M1

Material Type: FERT

Unit: EA

Σ
63

 Date

GR
 ST On
 On
 Vendor
Cust.

A...	Date	MRP e...	MRP element data	Rescheduling...	E...	Receipt/Reqmt	Available Qty	Pro...
	08/02/2023	Stock						0
	08/01/2023	IndReq	VSF				168-	168-
	08/09/2023	----->	End of Planning Tim...					
	08/09/2023	PldOrd	0000000423/STCK	08/01/2023	30	168		0 0001
	09/01/2023	PldOrd	0000000424/STCK			161		161 0001
	09/01/2023	IndReq	VSF			161-		0
	10/01/2023	PrdOrd	000001000002/PP01/Re	08/01/2023	10	142		142 0001
	10/01/2023	IndReq	VSF			142-		0

Step 9 – Receive Goods in Inventory

Posted goods receipt (500 EA each) for all raw/semi-finished components without reference, to enable production.

- **Material Document Number:** [شماره ی خودت]

Select Storage Location

Search

Standard

Semi-Fin. Goods

SF00

Unrestricted... 0.000 EA

Quality	0.000 EA
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Blocked	0.000 EA
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☐	Item	Material	Quantity / Unit			Plant	Storage Location		Stock Type	Special Stocks		
☐	01	TWNA1400	1	500,000	EA	1	Plant Dallas (DL00)	1	Semi-Fin. Goods (SFG0)	1	Unrestricted Use	None
☐	02	TFRF3400	1	500,000	EA	1	Plant Dallas (DL00)	1	Raw Materials (RM00)	1	Unrestricted Use	None
☐	03	DGAM1400	1	500,000	EA	1	Plant Dallas (DL00)	1	Raw Materials (RM00)	1	Unrestricted Use	None
☐	04	TFSK1400	1	500,000	EA	1	Plant Dallas (DL00)	1	Raw Materials (RM00)	1	Unrestricted Use	None
☐	05	TFRB1400	1	500,000	EA	1	Plant Dallas (DL00)	1	Raw Materials (RM00)	1	Unrestricted Use	None
☐	06	PEDL1400	1	500,000	EA	1	Plant Dallas (DL00)	1	Raw Materials (RM00)	1	Unrestricted Use	None
☐	07	CHAN1400	1	500,000	EA	1	Plant Dallas (DL00)	1	Raw Materials (RM00)	1	Unrestricted Use	None
☐	08	BRNT1400	1	500,000	EA	1	Plant Dallas (DL00)	1	Raw Materials (RM00)	1	Unrestricted Use	None
☐	09	WDOCC1400	1	500,000	EA	1	Plant Dallas (DL00)	1	Raw Materials (RM00)	1	Unrestricted Use	None
☐	10	PCKGL1400	1	500,000	EA	1	Plant Dallas (DL00)	1	Raw Materials (RM00)	1	Unrestricted Use	None

Step 10 — Issue Goods to Production Order

Issued all components to the production order (movement type 261).

- **Material Document Number:** [شماره ی خودت]

Goods issue — itemized component list

Step 11 — Review Production Order Status

Confirmed all component reservations were fully consumed (open quantity = 0).

Area of Responsibility (Work Center)

Step 12 — Confirm Production Completion

Entered final confirmation with full yield quantity and no scrap.

Confirmation — actual data entry

Step 13 — Review Production Order Status

Order status changed from Released to Confirmed; processing moved to “Move to storage.”

The screenshot shows the SAP Manage Production Orders interface. The order status is 'Confirmed' and the progress bar indicates 'Move to storage (01.10)'. The order is for 'Deluxe Touring Bike (red)' with a quantity of 142 EA.

Order	Material	Open Quantity	Status	Start	End	Progress of Operation	Issues
1000002	DXTR3400 Deluxe Touring Bike (red)	142 EA	Confirmed	Wed, Aug 2, 2023 07:02	Wed, Aug 2, 2023 08:02	Move to storage (01.10)	

Confirmation

Confirmation	Confirmation Count	Reversed	Reversal	Reversed Count	Operation	Confirmed By	Confirmed	Final Confirmation	Quantity	Yield
161	1	No	No			LEARN-400	Wed, Aug 2 08:02	Yes	142 EA	142 EA

Step 14 — Receive Goods from Production Order

Posted goods receipt into finished goods storage (FG00).

- **Material Document Number:** [شماره ی خودت]

The screenshot shows the SAP Items table for the production order. It displays the material 'Deluxe Touring Bike (red)' with a quantity of 142,000 EA, distributed to the plant 'Plant Dallas (DL00)' in the storage location 'Finished Goods (FG00)'.

Material	Open Quantity	Delivered	Distribution	Plant	Storage Location	Stock Type
Deluxe Touring Bike (red)	142,000 EA	142,000 EA	+	Plant Dallas (DL00)	Finished Goods (FG00)	Unrestricted Use

Goods receipt for production order — finished goods

Step 15 — Review Costs Assigned to Production Order

Reviewed target vs. actual costs and the Manufacturing Output Settlement Variance.

The screenshot shows the SAP Production Cost Analysis interface. It displays the cost details for the production order, including the total target cost, total actual cost, and the manufacturing output settlement variance.

GL Account	Total Target Cost	Total Acti Cost	Tgt/Act Cost/Var	Actual Quantity	Origin	Origin Activity Type Name
Group Name: 7520000 (Settle. Manuf Ou...	0.00 USD	-103,951.10 USD	-103,951.10 USD	-142 EA	DL00DXTR3400	
Group Name: Human Resources 8000000 (Labor)	0.00 USD	3,551.55 USD	3,551.55 USD	71.031 HR	NAPR1000LABOR	Labor hours
Group Name: Raw Materials						
5001000 (RM Consumpt E...	0.00 USD	9,940.00 USD	9,940.00 USD	142 EA	DL00BRKT1400	
5001000 (RM Consumpt E...	0.00 USD	1,420.00 USD	1,420.00 USD	142 EA	DL00CHAM1400	
5001000 (RM Consumpt E...	0.00 USD	10,650.00 USD	10,650.00 USD	142 EA	DL00DGAM1400	
5001000 (RM Consumpt E...	0.00 USD	497.00 USD	497.00 USD	142 EA	DL00PGKL1400	
5001000 (RM Consumpt E...	0.00 USD	6,390.00 USD	6,390.00 USD	142 EA	DL00PEDL1400	
5001000 (RM Consumpt E...	0.00 USD	28,400.00 USD	28,400.00 USD	142 EA	DL00TRFR3400	
5001000 (RM Consumpt E...	0.00 USD	3,550.00 USD	3,550.00 USD	142 EA	DL00TRHB1400	
5001000 (RM Consumpt E...	0.00 USD	7,100.00 USD	7,100.00 USD	142 EA	DL00TRSK1400	
5001000 (RM Consumpt E...	0.00 USD	142.00 USD	142.00 USD	142 EA	DL00WDOC1400	
	0.00 USD	-218.55 USD	-218.55 USD	1,491.631 *		

Question answered: How is the Manufacturing Output Settlement Variance calculated by the system?

[جواب خودت اینجا]

Step 16 — Settle Costs of Production Order

Ran actual settlement (test run → update run) and confirmed costs were fully settled to the material.

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SAP

Actual Settlement: Order Basic list

Search In: "Apps"

Menu  

Selection

Selection Parameters	Value	Name
Order	1000002	Deluxe Touring Bike (red)
Period	008	
Posting Period	008	
Fiscal Year	2023	
Processing Type	1	Automatic
Posting Date	08/31/2023	
Controlling Area	NA00	Global Bike North America
Currency	USD	United States Dollar
Value Date	08/31/2023	

Processing Options

Selection Parameters	Value
Execution Type	<u>Settlement Executed</u>
Processing Mode	<u>Test run</u>

Processing completed with no errors

Statistics

Processing Category	z	Number
<u>Settlement Executed</u>		1
No Change		
Not Relevant		
Inappropriate Status		
Error		

Select Report

Orders: Actual/Plan/Variance

Orders: Actual/Plan/Commitments

Orders: Drilldown by Partner

Orders: Accruals/Category

✓ Technical names on/off ✗

Processing Options

Selection Parameters	Value
Execution Type	<u>Settlement Executed</u>
Processing Mode	<u>Update run</u>

Processing completed with no errors

Key Concepts Applied

- **Master data hierarchy:** material master, routing, BOM as the foundation for planning
- **Sales & Operations Planning (SOP):** high-level forecasting via seasonal trend models
- **Demand Management:** disaggregation of product-group-level plans to material level
- **MRP net requirements logic:** comparing available quantity vs. requirements to trigger procurement proposals
- **Production order lifecycle:** planned order → production order → confirmation → goods receipt
- **Cost object controlling:** target/actual cost variance and settlement to the finished material

Tech Stack

- SAP S/4HANA 2022
- SAP Fiori 3.0
- Dataset: Global Bike (SAP University Alliances)

Notes

Student/company-specific identifiers (e.g., the ### three-digit suffix) have been omitted from this write-up to keep the repo generic and shareable.